GLM - Part I: Logistic Regression

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Book chapters

- **An Introduction to Statistical Learning (ISLR):** Chapter 4 (Classification). Key sections:
    - **4.2** Why Not Linear Regression?
    - **4.3** Logistic Regression (The Logistic Model, Estimating coefficients, Making Predictions, Multiple Logistic Regression)
- **The R Book (Crawley):** Chapter 13 (Generalized Linear Models). Key sections:
    - **Introduction to GLMs** (Error structures, Linear Predictors, and Link functions)
    - **Binomial Data** (Binomial Errors for grouped data)
    - **Binary Response Variables** (Modeling individual 0/1 outcomes)

GLM's - What for?

- A bigger class of linear models
- Most single-response linear models are GLM's
- Allows non-normally distributed response
    - Binary response
    - Binomial response
    - Count response
    - Skewed data - Gamma response
    - Multinomial response
    - And more…

Today's topics

- Motivating examples: why linear regression fails for binary data
- Logistic regression (binary or Binomial response)
- Estimation of parameters and Inference
- Application to data examples
- Binary classification revisited

Recall linear regression

From our earlier lectures, we have observed data with a continuous response y_i and predictors $x_{i1}, \ldots, x_{ip}$.

The multiple linear regression model is:

$$\mu(\boldsymbol{x}_i) = E(y_i) = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}$$

where the observations y_i are typically assumed to be normally distributed around the mean $\mu(\boldsymbol{x}_i)$ with some variance σ^2:

$$y_i \sim N(\mu(\boldsymbol{x}_i), \sigma^2)$$

What happens if the response variable y_i is **binary**, meaning it can only take values 0 or 1?

What if the response is binary?

Imagine we want to predict if a client will default on a loan ($y \in \{0, 1\}$) based on their account Balance ($x \in R$).

If we fit a simple linear regression model:

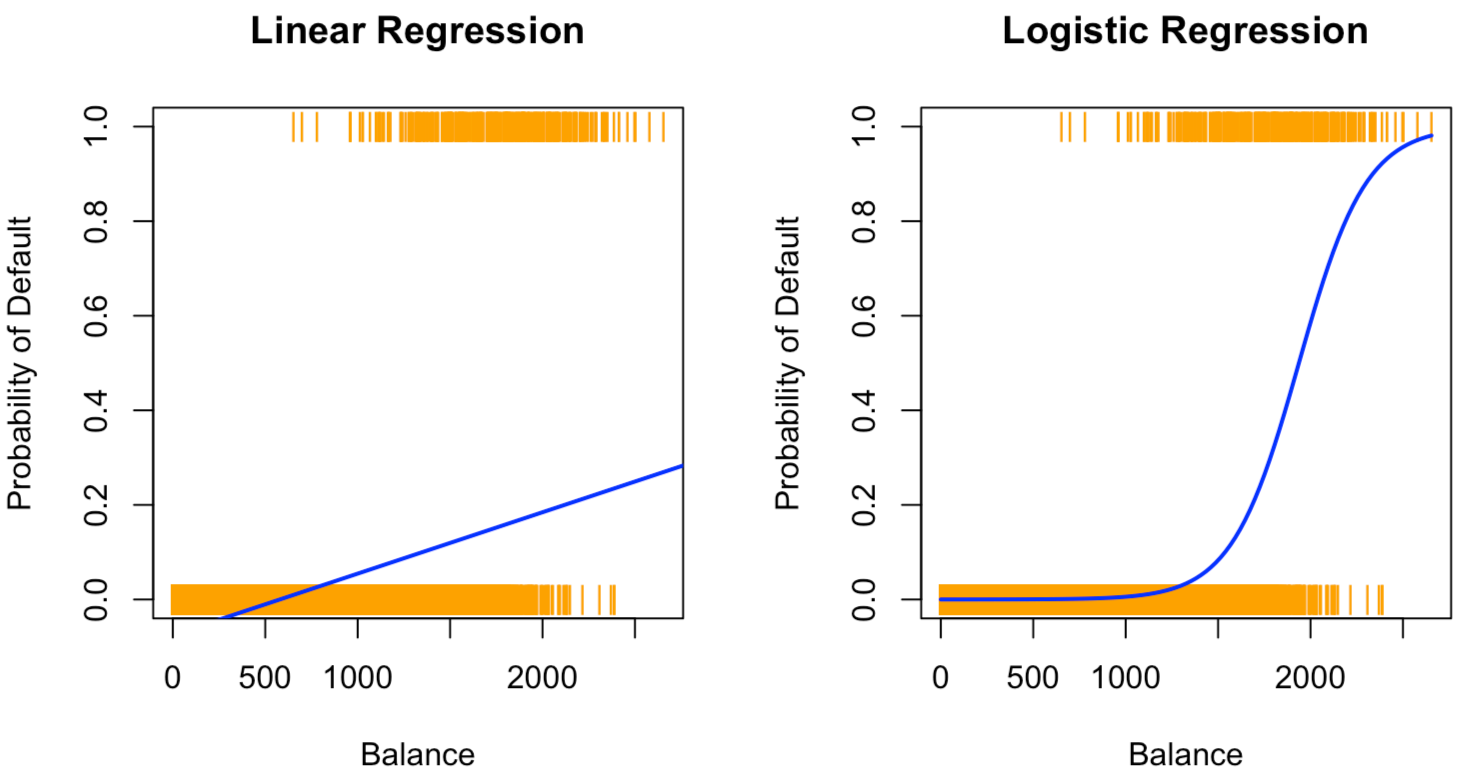
$$E(y_i) = \beta_0 + \beta_1 x_{i1}$$

We want the expected value $E(y_i)$ to represent the **probability** $P(y_i = 1)$, right? However, a straight line can predict probabilities less than 0 or greater than 1, which is not possible!

What if the response is binary? (Data)

```
##   default student   balance    income
## 1      No      No  729.5265 44361.625
## 2      No     Yes  817.1804 12106.135
## 3      No      No 1073.5492 31767.139
## 4      No      No  529.2506 35704.494
## 5      No      No  785.6559 38463.496
## 6      No     Yes  919.5885  7491.559
```

What if the response is binary? (Plot)



Modeling the Response

In Linear Regression, we assumed the continuous response y_i followed a **Normal Distribution** centered around its mean μ_i:

$$y_i \sim N(\mu_i, \sigma^2)$$

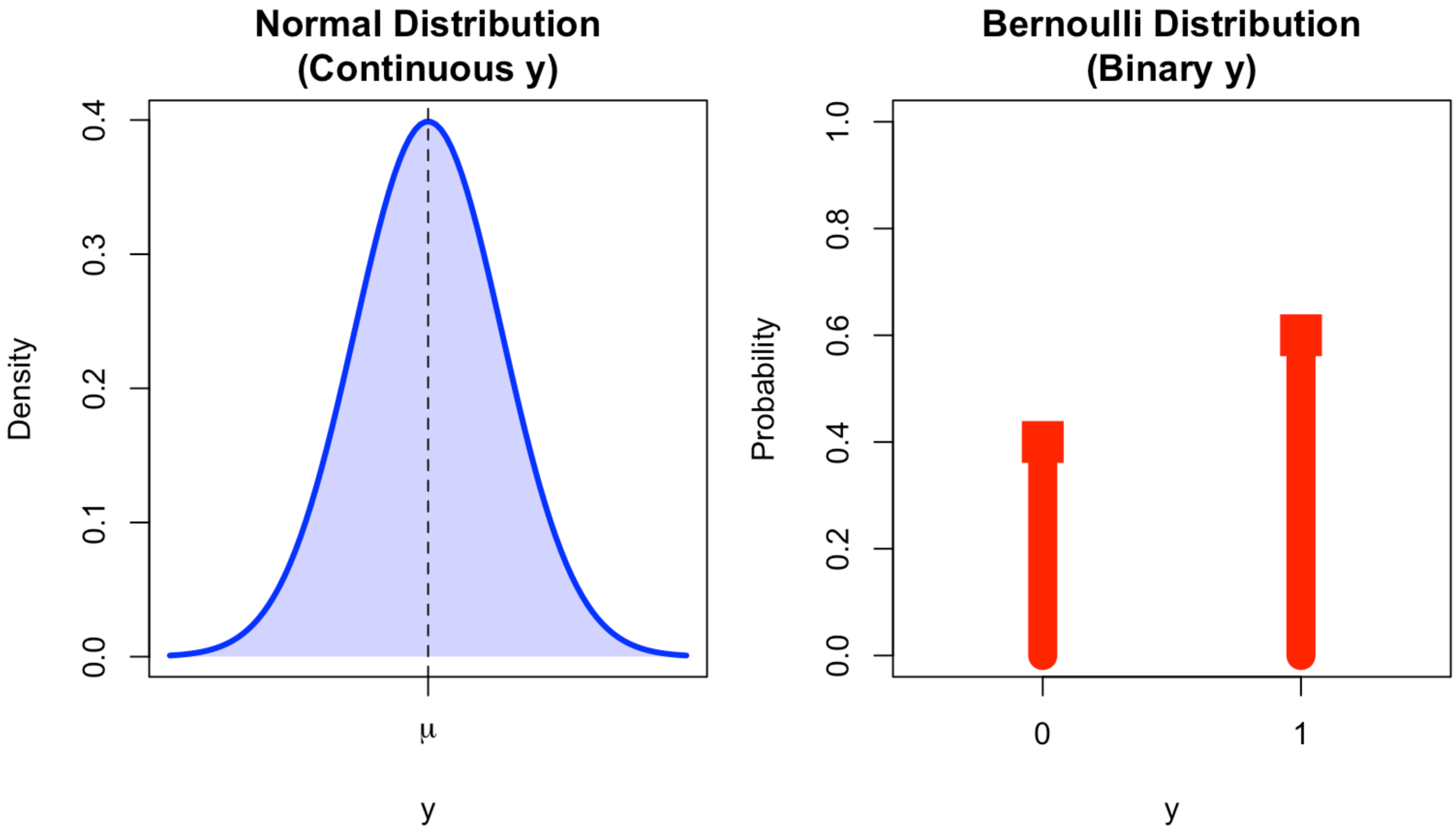
For a binary response, the data can only be 0 or 1. This perfectly matches the **Bernoulli Distribution**!

The Bernoulli distribution describes a single trial that results in a "success" ($y = 1$) with probability p_i, or "failure" ($y = 0$) with probability $1 - p_i$:

$$y_i \sim \text{Bernoulli}(p_i)$$

Our "mean" prediction is now simply the probability: $E(y_i) = p_i$.

Normal vs Bernoulli Distribution



Solution: Logistic regression

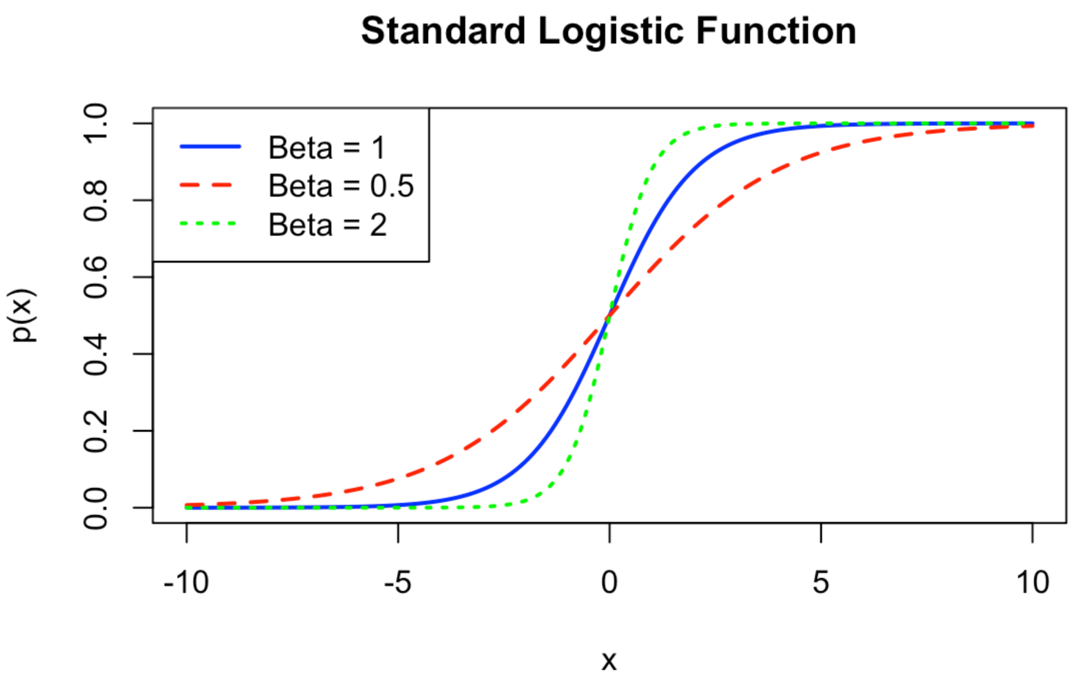
In logistic regression, we model the probability $p_i = p(\boldsymbol{x}_i) = P(y_i = 1|\boldsymbol{x}_i)$ directly, ensuring it stays between 0 and 1.

Instead of a straight line, we use the **logistic function** (or sigmoid function):

$$p(\boldsymbol{x}_i) = \frac{e^{\beta_0+\beta_1 x_{i1}+\cdots+\beta_p x_{ip}}}{1 + e^{\beta_0+\beta_1 x_{i1}+\cdots+\beta_p x_{ip}}}$$

This ensures that no matter what values the predictors x take, the predicted probability p_i is always theoretically constrained: $0 \leq p(\boldsymbol{x}_i) \leq 1$.

The Logistic Function Shape



Notice how it “squashes” the predictions to stay within 0 and 1.

Log-odds (Logit) connection

How does logistic regression connect to linear regression?

If we rearrange the logistic formula, we find that the **log-odds** is modeled linearly!

$$\text{Odds} = \frac{p(\boldsymbol{x}_i)}{1 - p(\boldsymbol{x}_i)} = \frac{P(y_i = 1|\boldsymbol{x}_i)}{P(y_i = 0|\boldsymbol{x}_i)}$$

The logarithm of the odds is called the **logit** function:

$$\log\left(\frac{p(\boldsymbol{x}_i)}{1 - p(\boldsymbol{x}_i)}\right) = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}$$

So, a generalized representation uses a *link function* — here, the logit link connects the mean (probability $p(\boldsymbol{x}_i)$) to the linear predictor: $\beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}$

Derivation of log odds from $p(\boldsymbol{x}_i)$

$$\log\left(\frac{p(\boldsymbol{x}_i)}{1 - p(\boldsymbol{x}_i)}\right) = \beta_0 + \beta_1 x_{i1}$$

Interpretation of Logistic Coefficients

In a linear regression model, a coefficient β_1 means an increase of 1 in x_1 leads to a β_1 increase in $E(y)$.

What does β_1 mean in Logistic regression?

Looking at the log-odds equation:

$$\log(\text{Odds}) = \beta_0 + \beta_1 x_1$$

If we increase x_1 by one unit:

- The **log-odds** increases by β_1.

- The **odds** are multiplied by e^{β_1}.

e^{β_1} is called the **Odds Ratio**.

Derivation of Interpretations

Looking at the log-odds equation:

$$\log(\text{Odds}) = \beta_0 + \beta_1 x_1$$

If we increase x_1 by one unit:

Parameter Estimation: Maximum Likelihood

In linear regression, we used the Least Squares method (which is equivalent to Maximum Likelihood for normally distributed errors) to find the best β coefficients.

For Logistic Regression, there is no analytical "formula" (like the least squares equations) for the coefficients.

Instead, the computer uses numerical algorithms to iteratively find the β values that maximize the **Likelihood function**.

$$L(\boldsymbol{\beta}) \rightarrow \max$$

Maximizing the likelihood means finding the parameter values that make our observed binary data *most probable*.

Likelihood Factorization

Since observations are independent, the total probability (Likelihood) of seeing our data is the product of individual probabilities.

For a Binomial/Bernoulli outcome $y_i \in \{0, 1\}$, the probability is p_i if $y_i = 1$, and $(1 - p_i)$ if $y_i = 0$.

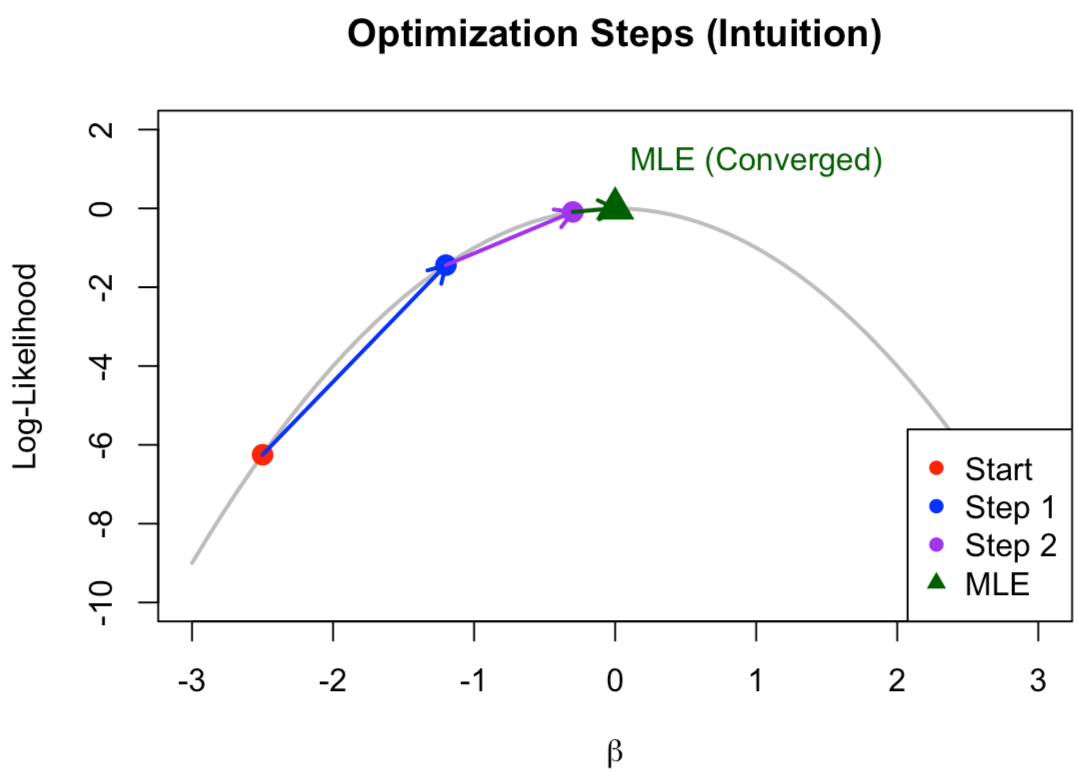
We can write this elegantly as:

$$L(\boldsymbol{\beta}) = \prod_{i=1}^{n} p_i^{y_i} (1 - p_i)^{1-y_i}$$

We want to find $\boldsymbol{\beta}$ that maximizes this product (or its sum of logs!).

How the computer finds the Maximum

Instead of a set formula, the computer takes “steps” uphill on the Log-Likelihood curve to find the peak!



The Wald Test

Just like linear regression, the maximum likelihood estimates $\widehat{\beta}_j$ have standard errors, $SE(\widehat{\beta}_j)$.

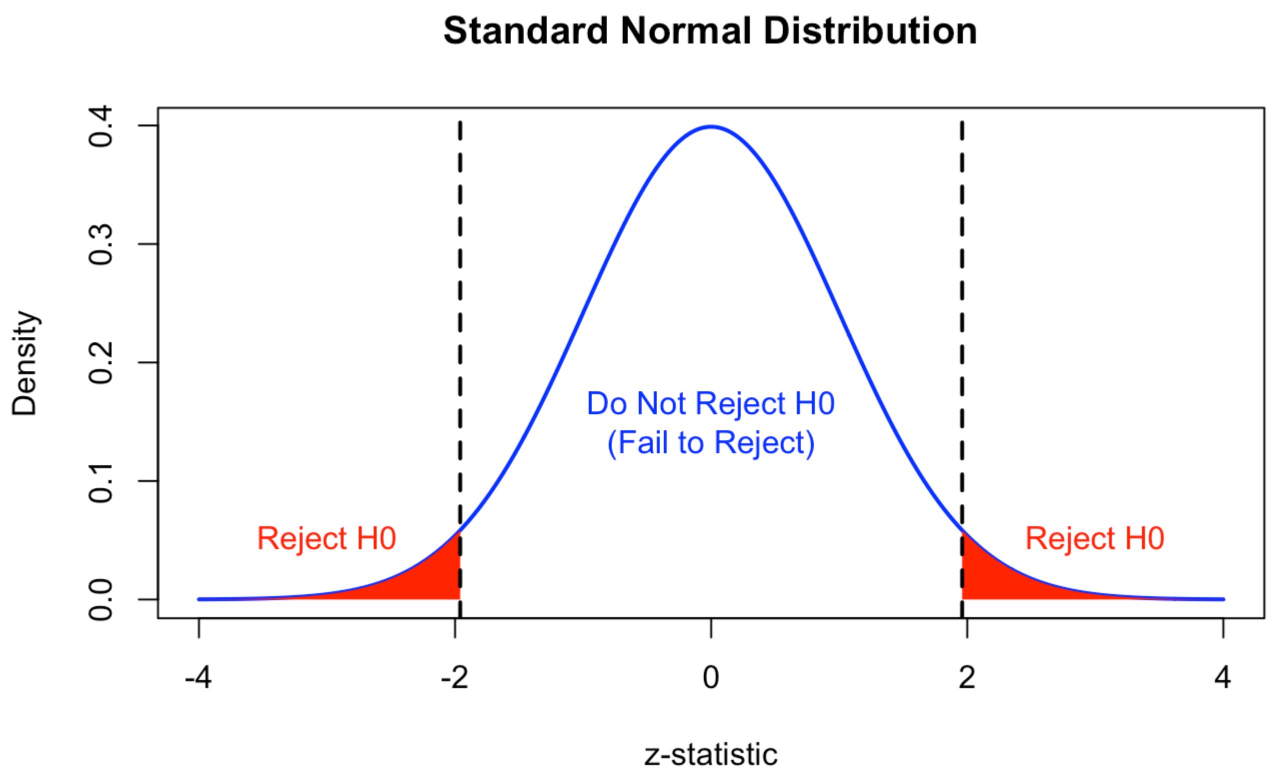
For large sample sizes, $\widehat{\beta}_j$ approximately follows a normal distribution.

We can test the hypothesis for individual coefficients ($H_0 : \beta_j = 0$ vs $H_A : \beta_j \neq 0$) using the **Wald test**:

$$z = \frac{\widehat{\beta}_j - 0}{SE(\widehat{\beta}_j)}$$

This statistic z follows a Standard Normal distribution $N(0, 1)$. Software (like R) will output these z-values and corresponding p-values.

Wald Test Decision Boundaries



Standard Errors from MLE Theory

How does the computer know the Standard Error (SE) of the estimates?

It uses the **curvature** of the log-likelihood peak!

- **Sharp peak:** We are very confident in the estimate. SE is small.
- **Flat peak:** There is a lot of uncertainty. SE is large.

The mathematical term is the **Fisher Information** (I), which measures the steepness (second derivative) of the peak.

$$SE(\widehat{\beta}) = \sqrt{\frac{1}{I(\widehat{\beta})}}$$

So, sharp curvature = large Information = small Standard Error!

The Motivation: We want our proposed predictors to significantly reduce the Lack of Fit. Therefore, we want the **Residual Deviance** to be much smaller than the **Null Deviance**!

Likelihood Ratio Test (Deviance)

To test if a subset of predictors as a whole is useful, we compare nested models using the **Likelihood Ratio Test (LRT)**.

This is analogous to comparing the Sum of Squared Errors (SSE) in ANOVA for linear models. Instead of SSE, we use **Deviance:** $-2 \log L(\boldsymbol{\beta})$.

- The Null Model contains only the intercept β_0.
- The Proposed Model contains the predictors.

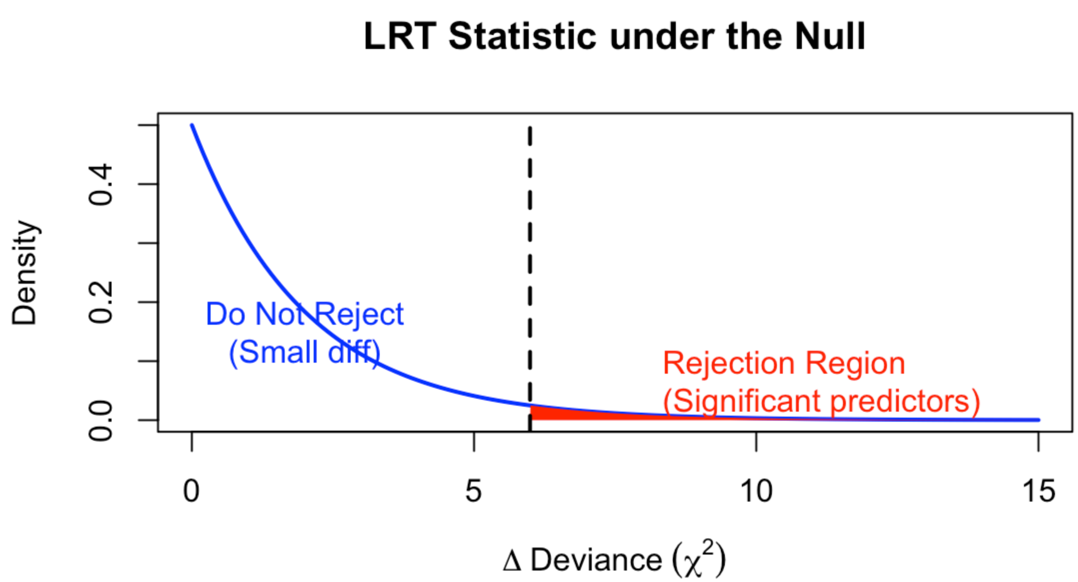
The test compares the Deviance of the models, and the difference in Deviances ($\Delta\text{Deviance}$) follows a χ^2 (Chi-square) distribution based on the number of additional predictors. Its degrees of freedom (df) is:

$$df = (\text{parameters in Proposed}) - (\text{parameters in Null})$$

Likelihood Ratio Test (Intuition)

The Deviance compares the lack of fit of two models. If the Proposed Model is **much better**, its deviance is much lower, making the difference $\Delta\text{Deviance}$ large!

- **Small difference:** The extra predictors don't do much.
- **Large difference:** Pushes into the right tail (Rejection Region). The predictors are useful!



Assumptions of Logistic Regression

Like linear models, GLMs have underlying assumptions we must check:

1. **Independence:** Observations y_i are independent of each other.
2. **Linearity in the Logit:** The log-odds (logit) relates *linearly* to the continuous predictors.
3. **No Multicollinearity:** Predictors shouldn't be too perfectly correlated.
4. **No Extreme Outliers:** No single observation should unilaterally drive the model.
5. **No Overdispersion** (for Binomial data): The observed variance shouldn't be higher than what the Binomial theory assumes.

Example 1: Bird species on islands

Let’s test this in R! We look at the probability of a bird species being present on an island.

```
##   incidence  area isolation
## 1         1 7.928     3.317
## 2         0 1.925     7.554
## 3         1 2.045     5.883
## 4         0 4.781     5.932
## 5         0 1.536     5.308
## 6         1 7.369     4.934
```

- incidence = 1 if the species is present, 0 if absent
- area = Area of the island (km^2)
- isolation = Distance to the mainland (km)

Question: How does the probability of presence depend on the area?

Fitting the Logistic Model in R

We use the `glm()` function, specifying `family=binomial(logit)` for logistic regression. `mixlm` package is used over the standard base R because it provides specific expanded summaries and visualizations in the context of the course, or we can just use the base `glm`.

```
library(mixlm)
GLM.1 <- glm(incidence ~ area, family=binomial(logit), data=island)
summary(GLM.1)
```

The output

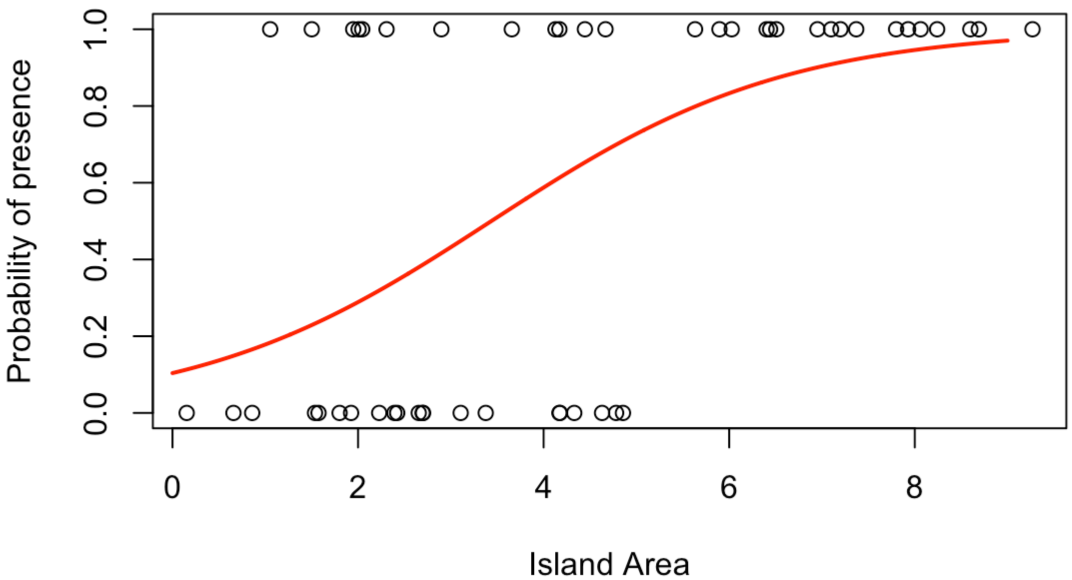
```
##
## Call:
## glm(formula = incidence ~ area, family = binomial(logit), data = island)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -2.1554     0.7545  -2.857 0.004278 **
## area          0.6272     0.1861   3.370 0.000753 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 68.029  on 49  degrees of freedom
## Residual deviance: 50.172  on 48  degrees of freedom
## AIC: 54.172
##
## Number of Fisher Scoring iterations: 5
```

Interpreting the output

- Intercept $\widehat{\beta}_0 = -2.16$
- Area coefficient $\widehat{\beta}_{\text{area}} = 0.63$. The positive coefficient means larger area increases the odds of bird presence!
- The **p-value** is < 0.001, found using the Wald test (z value = 3.370). The coefficient is significantly different from 0.
- **Odds ratio**: $e^{0.63} \approx 1.872$. For every 1 km^2 increase in area, the odds of the species being present on the island increase by about 87.2%.

Graphical Representation of the Fit

```
xv <- seq(0,9,0.01)
yv <- predict(GLM.1, list(area=xv), type="response")
plot(island$area, island$incidence, ylab="Probability of presence", xlab="Island Area")
lines(xv, yv, col="red", lwd=2)
```



Goodness of fit (LRT): Area against the `Null`

`anova()` function with `test="LR"` or `test="Chisq"`.

```
anova(GLM.1, test='LR')
```

```
## Analysis of Deviance Table
##
## Model: binomial, link: logit
##
## Response: incidence
##
## Terms added sequentially (first to last)
##
##
##      Df Deviance Resid. Df Resid. Dev  Pr(>Chi)
## NULL                    49     68.029
## area  1   17.857        48     50.172 2.382e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Goodness of fit (LRT) cont.

The Likelihood Ratio test from the previous output shows a significant result: p-value<0.05.

This indicates that adding the `area` predictor significantly improves the fit of the model compared to a simple baseline model with just the intercept.

Expanding the model

What if we also include isolation?

```
##
## Call:
## glm(formula = incidence ~ area + isolation, family = binomial(logit),
##     data = island)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)   6.6417     2.9218   2.273  0.02302 *
## area          0.5807     0.2478   2.344  0.01909 *
## isolation    -1.3719     0.4769  -2.877  0.00401 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 68.029  on 49  degrees of freedom
## Residual deviance: 28.402  on 47  degrees of freedom
## AIC: 34.402
##
## Number of Fisher Scoring iterations: 6
```

Goodness of fit (LRT)

Comparing the models progressively:

```
anova(GLM.2, test='LR')
```

```
## Analysis of Deviance Table
##
## Model: binomial, link: logit
##
## Response: incidence
##
## Terms added sequentially (first to last)
##
##
##           Df Deviance Resid. Df Resid. Dev  Pr(>Chi)
## NULL                         49     68.029
## area       1   17.857        48     50.172 2.382e-05 ***
## isolation  1   21.770        47     28.402 3.073e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Both variables give significant drops in deviance.

Comparing Models: The AIC

The **Akaike's Information Criterion (AIC)** is commonly used to compare models. It is defined as:

$$-2 \times \log L(\boldsymbol{\beta}) + 2 \times p$$

where $\log L(\boldsymbol{\beta})$ is the log-likelihood of the model and p is the number of estimated parameters. It penalizes complex models. **Smaller is better**.

Comparing Models: The AIC (Results)

```r
AIC(GLM.1) # Area only
```

```
## [1] 54.17231
```

```r
AIC(GLM.2) # Area + Isolation
```

```
## [1] 34.40216
```

Adding isolation improves the model significantly, leading to a lower AIC!

Comparing Models: BIC

Along with AIC, we also use the **Bayesian Information Criterion (BIC)**:

$$-2 \times \log L(\boldsymbol{\beta}) + \log(n) \times p$$

BIC penalizes complex models even more harshly than AIC because $\log(n)$ is usually greater than 2. Again, **smaller is better**.

```
BIC(GLM.1) # Area only
```

```
## [1] 57.99636
```

```
BIC(GLM.2) # Area + Isolation
```

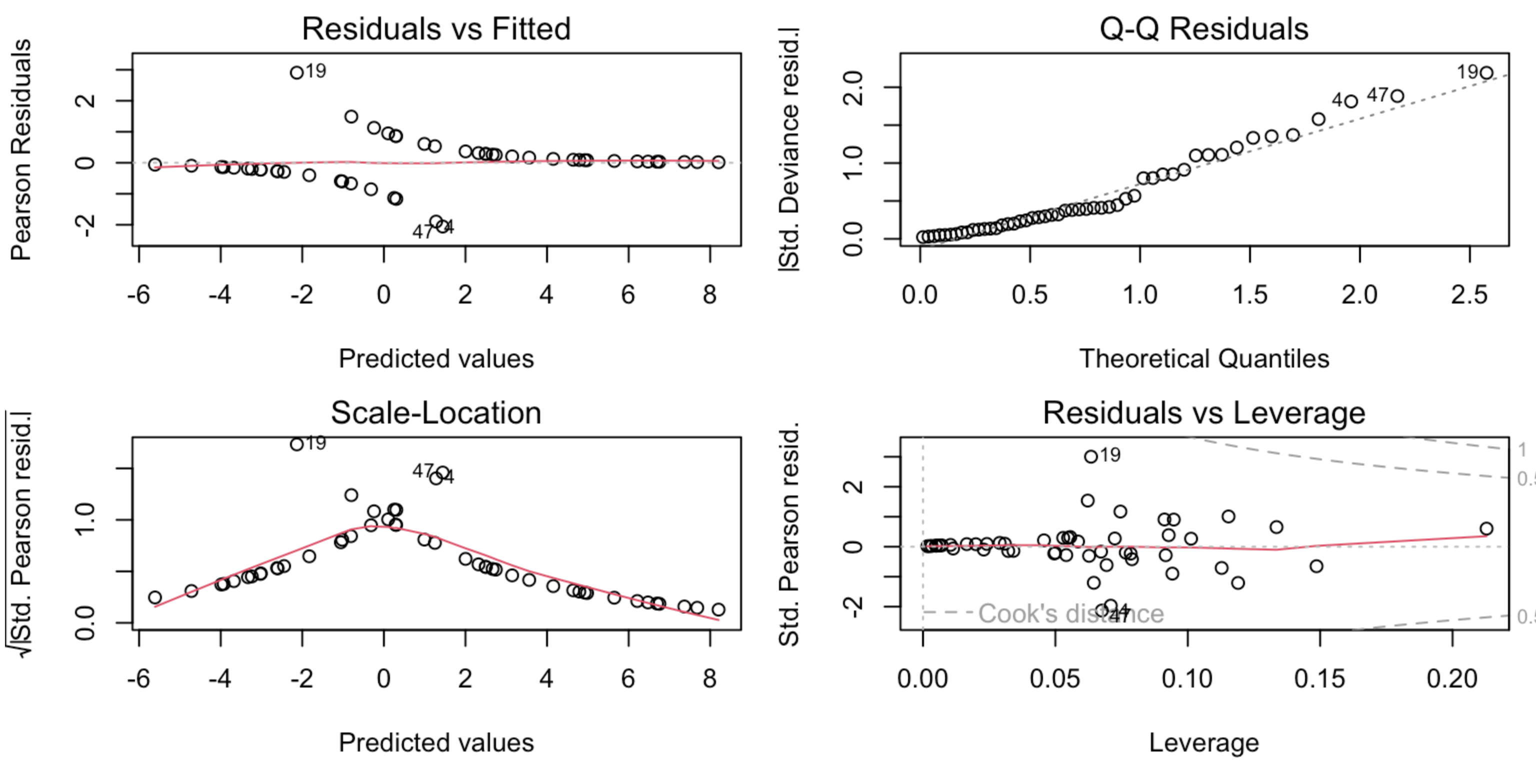
```
## [1] 40.13823
```

Both AIC and BIC agree: adding isolation improves the model!

Checking the Bird Model (Diagnostics)

Let's verify our assumptions by generating all standard diagnostic plots for `GLM.2`:

```
par(mfrow=c(2,2), mar=c(4,4,2,1))
plot(GLM.2)
```



Binomial response

Sometimes, rather than tracking individual 0s and 1s, we have binomial data.

$$y_i \text{ successes out of } n_i \text{ trials}$$

For $i = 1, \dots, n$ where y_i is the number of 'successes' among n_i total experiments in group i.

We assume $y_i \sim \text{Binomial}(n_i, p_i)$.

Binomial response (cont.)

Theory tells us:

$$E(y_i) = p_i$$

$$Var(y_i) = \frac{p_i(1 - p_i)}{n_i}$$

We can use logistic regression to model p_i using predictors, functioning precisely the same way!

Example 2: The insect sex data

The sex counts in insects (proportion males) depending on population density:

```
##   density females males
## 1       1       1     0
## 2       4       3     1
## 3      10       7     3
## 4      22      18     4
## 5      55      22    33
## 6     121      41    80
## 7     210      52   158
## 8     444      79   365
```

Binomial regression in R

In RStudio, the response for Binomial grouped data must be an array of `successes` and `failures`.

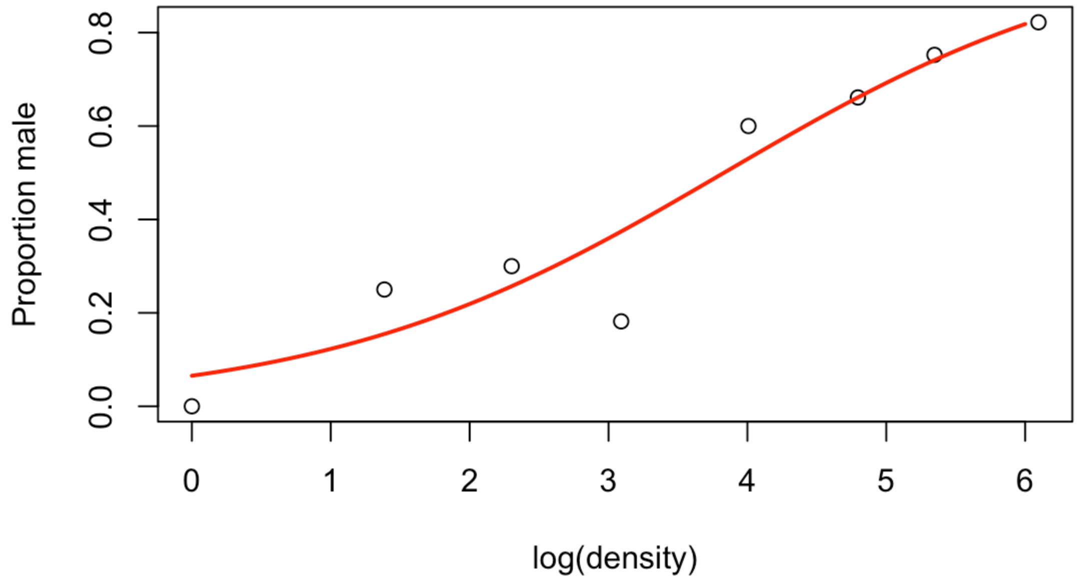
In the model specification, use `cbind(succesess, failures)` as the response variable.

```
GLM.4 <- glm(cbind(males, females) ~ log(density),
             family=binomial(logit), data=numbers)
```

Insect sex-ratio model summary

```
##
## Call:
## glm(formula = cbind(males, females) ~ log(density), family = binomial(logit),
##     data = numbers)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -2.65927    0.48758  -5.454 4.92e-08 ***
## log(density)  0.69410    0.09056   7.665 1.80e-14 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 71.1593  on 7  degrees of freedom
## Residual deviance:  5.6739  on 6  degrees of freedom
## AIC: 38.201
##
## Number of Fisher Scoring iterations: 4
```

Insect sex-ratio plot



Binary classification revisited

Remember from Multivariate Analysis: Assume we have two groups (classes), c_1 and c_2. By Bayes rule, the posterior probability of class 2 given observed variables is:

$$P(c_2|x_1, \dots, x_k) \propto f(x_1, \dots, x_k|c_2)P(c_2)$$

Logistic regression gives an alternative, **direct** method to estimate this posterior probability!

Define $y = 1$ for c_2 and $y = 0$ for c_1. Then logistic regression directly estimates:

$$P(c_2|x_1, \dots, x_k) = P(y = 1|x_1, \dots, x_k) = \frac{e^{\beta_0+\beta_1 x_1+\dots}}{1 + e^{\beta_0+\beta_1 x_1+\dots}}$$

We classify to class 2 if the estimated $P(y = 1|x_1, \dots, x_k) > 0.5$.

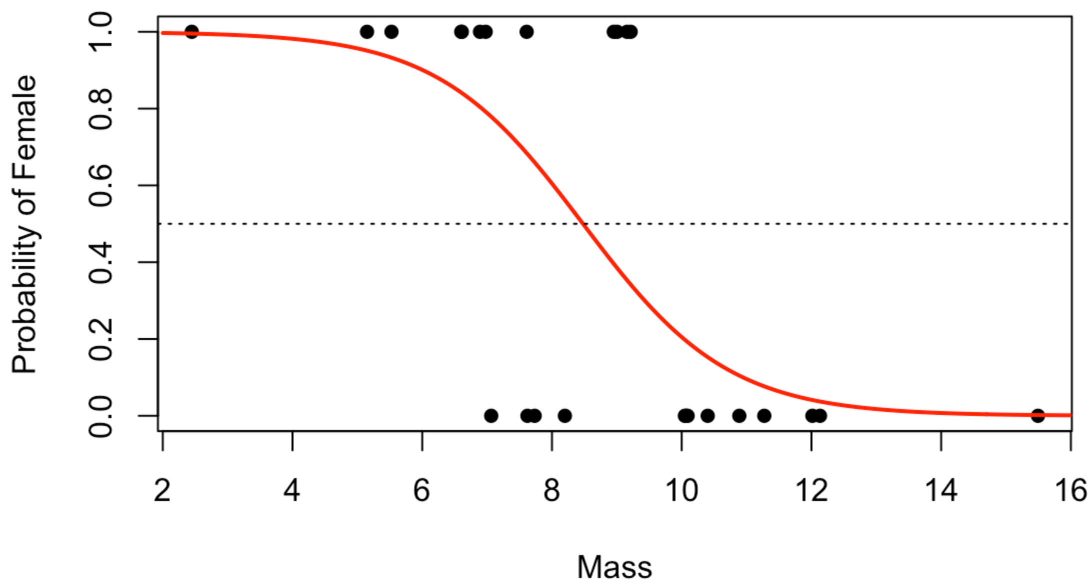
Example 3: Lizards

Can we discriminate the Sex of Lizards based on Mass? Let $y = 1$ if female, $y = 0$ if male.

```
##              Estimate Std. Error   z value   Pr(>|z|)
## (Intercept)  7.549321  3.0619963  2.465490 0.01368261
## Mass        -0.890471  0.3551573 -2.507258 0.01216720
```

For every 1 unit increase in Mass, the log-odds of being female changes by -0.89. Thus heavier lizards are more likely to be male!

Lizards: Fitted line plot



Classify a new lizard

Assume we have measured a new lizard with mass 5g. What is the most probable gender for this lizard given our estimated model?

```
new_lizard_data <- data.frame(Mass = 5)
ypred <- predict(model, new_lizard_data, type="response")
ypred
```

```
##         1
## 0.9567674
```

This returns the fitted estimated probability of being female ($y = 1$).

Since $0.957 > 0.5$, we classify the lizard to the female class. Hence, a lizard of Mass 5g is most probably a female.

Lizards: Variable Selection

Let's include all predictors (`Mass`, `SVL` - Snout-vent length, `HLS` - Hind limb span) and perform stepwise selection!

```
full_model <- glm(y ~ Mass + SVL + HLS, family=binomial, data=Lizard)
selected_model <- step(full_model, direction="both", trace=0,k = 2)
summary(selected_model)$coef
```

```
##               Estimate Std. Error      z value  Pr(>|z|)
## (Intercept) 1764.01091 1128477.11  0.001563178 0.9987528
## HLS          -13.78172    8820.73 -0.001562424 0.9987534
```

The stepwise method selected only `HLS` as the best predictor (lowest AIC).

Lizards: Out of sample evaluation

Let’s measure how well our model really works by splitting data into a **training set** and a **test set**.

```
set.seed(42)
train_idx <- sample(1:nrow(Lizard), size = 0.6 * nrow(Lizard))
train_data <- Lizard[train_idx, ]
test_data <- Lizard[-train_idx, ]

# Fit on training data
train_model <- glm(ifelse(Sex=="m", 0, 1) ~ HLS,
                   family=binomial, data=train_data)
```

Lizards: Training sample accuracy

First, let's evaluate the model on the **training data**:

```
train_probs <- predict(train_model, train_data, type="response")
train_preds <- ifelse(train_probs > 0.5, 1, 0)
train_actuals <- ifelse(train_data$Sex=="m", 0, 1)

# Training Confusion Matrix
table(Predicted = train_preds, Actual = train_actuals)
```

```
##          Actual
## Predicted 0 1
##         0 7 0
##         1 0 8
```

Training Accuracy: 1

Lizards: Testing sample accuracy

```
test_probs <- predict(train_model, test_data, type="response")
test_preds <- ifelse(test_probs > 0.5, 1, 0)
test_actuals <- ifelse(test_data$Sex=="m", 0, 1)

# Test Confusion Matrix
table(Predicted = test_preds, Actual = test_actuals)
```

```
##          Actual
## Predicted 0 1
##         0 6 0
##         1 0 4
```

Test Accuracy: 1